

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Computer Applications (BCA)

Semester-I University Examination March-2018

CA102 Programming Principles and Logic Development

Date: 27.03.2018, Tuesday Time: 10:00 am to 01:00 pm Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-I

Q.1 Attempt the following.

[11]

1. Define: Computer System
2. Define: Modularization.
3. The two most commonly used tools for planning a program's logic are _____.
(a) Flowchart and Pseudocode (b) ASCII and EBCDIC
(c) Java and Visual Basic (d) Word Processors and Spreadsheet
4. _____ is an assignment operator.
(a) = (b) - (c) + (d) *
5. A message that asks the user to enter the value is called _____.
(a) Comment (b) Prompt (c) Echo (d) None of above
6. What is the name for the process of paying attention to the important properties while ignoring non-essential details?
(a) Extraction (b) Extinction (c) Abstraction (d) Modularization
7. _____ is the non-executing statement that programmers place within their code to explain program statements in English.
(a) Comment (b) Pseudocode (c) Module (d) User Documentation
8. Define: Boolean Expression
9. Symbols such as > and < are known as _____ operators.
(a) Arithmetic (b) Sequential (c) Relational Comparison (d) Scripting Accuracy
10. Define: OR operator
11. In a flowchart, _____ is a process symbol.
(a) Parallelogram (b) Rectangle (c) Circle (d) None of above

Q.2 [A] Draw flowchart and write pseudocode for a cell phone company that charges basic monthly service bill \$30 to all customers. An additional \$20 is billed to customers who make more than 100 calls that last for a total of more than 500 minutes. **[06]**

[B] Draw a flowchart and write a pseudocode to represent the logic of a program that allows the user to enter two values. The program outputs all the basic mathematical operations on that two values. **[06]**

OR

Q.2 [A] Explain features of good program design in detail. **[06]**
[B] Explain program development life cycle in detail. **[06]**

[P.T.O]

Q.3 Answer following questions in detail (Any three) [12]

1. What is called operators precedence? Explain in detail.
2. Explain advantages of modularization.
3. What is called Truth Table? Explain with example.
4. Explain in detail relational comparison operators with examples.
5. What is Pseudocode? Explain with example. What is Flowchart? Explain with example.

SECTION-II

Q.4 Attempt the following. [11]

1. Define: Loop.
2. Define: Counter
3. When you _____, you make sure that data items are of correct type and fall within the correct range.
(a) Validate data (b) Employ Offensive programming
(c) Use Object Orientation (d) Count loop iterations
4. Each variable in an array must have the same _____.
(a) Data Type (b) Subscript (c) Value (d) Memory Location
5. The subscript of an array are always _____.
(a) Integers (b) Fractions (c) Characters (d) None of above
6. A _____ is a variable that you set to indicate whether some event has occurred.
(a) Subscript (b) Banner (c) Counter (d) Flag
7. Define: Parallel Arrays
8. Define: Computer File
9. Define: Volatile Memory
10. Which of the following is true about data hierarchy?
(a) Files contain records (b) Characters contain fields
(c) Fields contain files (d) Fields contain records
11. Define: Master File and Transaction File.

- Q.5 [A] Draw a flowchart and write a pseudocode to represent the logic of a program that allows the user to enter marks of three subjects out of 100 for five students. The program adds the marks of three subjects and finds the total and percentages. If marks in any subject are less than 35, Grade of the student is "F", otherwise grade is calculated based on percentage as given in the table. The program outputs the grade of a student. [06]**

Percentage	Grade
≥ 70	A
≥ 60 and < 70	B
≥ 50 and < 60	C
≥ 35 and < 50	D
< 35	F

- [B] Design the logic and write pseudocode for a program that outputs every number from 1 to 10. [06]**

OR

- Q.5 [A] Explain file operations in detail. [06]**
[B] Explain common loop applications in detail. [06]

Q.6 Answer following questions in detail (Any three)

[12]

1. What is meant by incrementing or decrementing? Explain with the suitable examples.
 2. Describe briefly how to avoid common loop mistakes?
 3. Define an Array. How arrays occupy computer memory?.
 4. What is a computer file? Explain its categories in detail.
 5. Write characteristics of a file.
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